

Beyond Benchmarks: What Might User-Centered Evaluation of Agentic AI Look Like?

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1. WHY DO WE NEED USER-CENTERED EVAL?

Agentic AI systems are rapidly moving from research prototypes to deployed user products. These systems shift interaction from direct manipulation [17] to *delegation* [13]: users express intent, while the agent determines how to act, when to ask for clarification, and how to recover from errors.

This shift exposes a gap in how we evaluate agentic systems. Agent quality is primarily measured through benchmarks, scripted evaluation suites, such as SWE-bench [8], WebArena [22], and AgentBench [11] that reduce performance to task success rates. While benchmarks have driven rapid progress, recent work has raised concerns about their validity. Trivial do-nothing agents can pass 38% of tasks on τ -bench, brittle string matching inflates WebArena scores, and weak baselines achieve non-trivial success across multiple widely used suites [9, 23]. Fundamentally, these evaluations treat agents in isolation, decoupled from the humans who must understand, monitor, delegate, and recover from their actions.

We identify four gaps that emerge when moving from benchmark-centric to user-centered agentic system evaluation:

(A) Agent success $\neq$ user success. An agent may complete all (sub)tasks correctly yet still fail the user through misaligned preferences and values, expectation gap, poor user experience, and lack of communication or intervention mechanisms [1, 20].

(B) The collaboration process is invisible. Benchmarks focus on quantitative measurable outcomes, but agent use is fundamentally process-heavy: when user goals evolves, when trust develops, and when delegation feels appropriate remain unaddressed by existing evaluation frameworks [6, 16].

(C) Trust and delegation are dynamic. Users continuously calibrate when to delegate decisions. Prior work shows that auto-approve rates can increase with experience as users learn to better estimate system reliability and the expected cost of errors versus intervention [2, 7], yet most evaluations treat trust as static or external to performance.

(D) Long-term effects remain unclear. Sustained use may reshape user behavior, skills, dependency, with downstream impacts on well-being and social dynamics. A system that performs well on benchmarks today may still be a net negative if it erodes user capability over time, yet such effects are not well captured in short-horizon evaluations [10, 12].

Wobbrock’s taxonomy of HCI research contributions [19] helps map the user-centric evaluation landscape for agentic systems: artifact contributions (building increasingly capable and

use-case-centric agents), dataset contributions (human datasets or benchmarks that define evaluation suites) dominate, empirical work studying how users actually interact with agents in the wild, and methodological work developing process-aware evaluation frameworks remain underrepresented.

2. TWO CASES WHERE AGENT SYSTEMS RAISE NEW EVALUATION CHALLENGES

To make these gaps concrete, we present two cases that illustrate evaluation challenges benchmarks cannot capture.

2.1. Case 1: Living with an Agent Over Time

Onboarding is nonlinear. Unlike traditional software, agentic systems do not expose functionality through static interfaces. Users coming from distinct backgrounds and technological experiences need to discover capabilities and limits through use or failure. As agents behavior varies with context, tools, and model updates, users continuously revise their usage, perception, and mental model of the system. Current agentic systems are also predominantly chat-based, which shape user perception of agent behavior but are rarely considered in evaluation [5, 14]. **Errors are subtle and compounding.** Agent failures are often delayed and distributed across steps. A single incorrect action may propagate through a long chain before becoming visible. Recovery is rarely local and may require re-executing entire workflows [3]. **Sustained use reshapes the user.** Over time, users develop new needs, expectations, prompts, habits, and skills – or abandon the system entirely. We lack longitudinal works for understanding appropriation, adaptation, and (non-)use of agentic systems [12].

2.2. Case 2: When the Agent Acts in Your Task

Proactivity vs. calm. There is a tension between “always proactive” agents and calm agents that remain in the background until needed [18]. Deciding when and how to interrupt, act, or defer is central to user experience but remains poorly understood [15, 21]. **Offloading and reliance.** Delegation depends on task stakes, familiarity, and experience, and evolves over time. This raises an agency paradox: users may value oversight in principle but delegate in practice [4]. How can we measure this gap empirically? **The trust-delegation landscape.** The goal is calibrated trust—neither undertrust (over-monitoring, underuse) nor overtrust (automation bias, skill loss). Should we be tracking trust trajectories rather than static ratings? What signals actually help users achieve calibration?

References

- [1] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, et al. Guidelines for human-ai interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*, pages 1–13, 2019.
- [2] Anthropic. Measuring AI agent autonomy in practice.
- [3] G. Bansal, J. W. Vaughan, S. Amershi, E. Horvitz, A. Fourney, H. Mozannar, V. Dibia, and D. S. Weld. Challenges in human-agent communication. *arXiv preprint arXiv:2412.10380*, 2024.
- [4] D. Bennett, O. Metatla, A. Roudaut, and E. D. Mekler. How does hci understand human agency and autonomy? In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–18, 2023.
- [5] A. Chan, C. Ezell, M. Kaufmann, K. Wei, L. Hammond, H. Bradley, E. Bluemke, N. Rajkumar, D. Krueger, N. Kolt, L. Heim, and M. Anderljung. Visibility into ai agents. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '24, page 958–973, New York, NY, USA, 2024. Association for Computing Machinery.
- [6] K. K. Feng, K. Pu, M. Latzke, T. August, P. Siangliulue, J. Bragg, D. S. Weld, A. X. Zhang, and J. C. Chang. Cocoa: Co-planning and co-execution with ai agents. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, pages 1–23, 2026.
- [7] E. Horvitz. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '99, page 159–166, New York, NY, USA, 1999. Association for Computing Machinery.
- [8] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. R. Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024.
- [9] S. Kapoor, B. Stroebl, Z. S. Siegel, N. Nadgir, and A. Narayanan. Ai agents that matter. *arXiv preprint arXiv:2407.01502*, 2024.
- [10] M. Kjærup, M. B. Skov, P. A. Nielsen, J. Kjeldskov, J. Gerken, and H. Reiterer. *Longitudinal studies in HCI research: a review of CHI publications from 1982–2019*. Springer, 2021.
- [11] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, et al. Agentbench: Evaluating llms as agents. *arXiv preprint arXiv:2308.03688*, 2023.
- [12] T. Long, S. Wang, E. Fabre, T. Wang, A. Sathya, J. Wu, S. D. Petridis, D. Li, T. Chakrabarty, Y. Jiang, J. Li, T. Tseng, K. Nakagaki, Q. Yang, N. Martelaro, J. V. Nickerson, and L. B. Chilton. Facilitating longitudinal interaction studies of ai systems. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, UIST Adjunct '25, New York, NY, USA, 2025. Association for Computing Machinery.
- [13] T. Long, X. Zhang, S. Wang, Z. Yu, and L. B. Chilton. Doubleagents: Human-agent alignment in a socially embedded workflow, 2026.
- [14] J. Lu, B. Pan, J. Chen, Y. Feng, J. Hu, Y. Peng, and W. Chen. Agentlens: Visual analysis for agent behaviors in llm-based autonomous systems. *IEEE Transactions on Visualization and Computer Graphics*, 31(8):4182–4197, 2025.
- [15] Y.-H. Peng, D. Li, J. P. Bigham, and A. Pavel. Morae: Proactively pausing ui agents for user choices. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, UIST '25, New York, NY, USA, 2025. Association for Computing Machinery.
- [16] R. Sahni, J. Ma, K. Sreedhar, and L. B. Chilton. Agentdynex: Nudging the mechanics and dynamics of multi-agent simulations. 2025.
- [17] Shneiderman. Direct manipulation: A step beyond programming languages. *Computer*, 16(8):57–69, 1983.
- [18] M. Weiser. The future of ubiquitous computing on campus. *Commun. ACM*, 41(1):41–42, jan 1998.
- [19] J. O. Wobbrock and J. A. Kientz. Research contributions in human-computer interaction. *Interactions*, 23(3):38–44, Apr. 2016.
- [20] Q. Yang, A. Steinfeld, C. Rosé, and J. Zimmerman. Re-examining whether, why, and how human-ai interaction is uniquely difficult to design. In *Proceedings of the 2020 chi conference on human factors in computing systems*, pages 1–13, 2020.
- [21] N. Zargham, L. Reicherts, M. Bonfert, S. T. Voelkel, J. Schoening, R. Malaka, and Y. Rogers. Understanding circumstances for desirable proactive behaviour of voice assistants: The proactivity dilemma. In *Proceedings of the 4th Conference on Conversational User Interfaces*, CUI '22, New York, NY, USA, 2022. Association for Computing Machinery.
- [22] S. Zhou, F. F. Xu, H. Zhu, X. Zhou, R. Lo, A. Sridhar, X. Cheng, T. Ou, Y. Bisk, D. Fried, et al. Webarena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023.
- [23] Y. Zhu, T. Jin, Y. Pruksachatkun, A. Zhang, S. Liu, S. Cui, S. Kapoor, S. Longpre, K. Meng, R. Weiss, et al. Establishing best practices for building rigorous agentic benchmarks. *arXiv preprint arXiv:2507.02825*, 2025.